BSBS • SYSTEM DEVELOPMENT PROJECT

# A DIGITAL SOLUTION FOR REPORTING AND TRACKING INFRASTRUCTURE ISSUES IN EDUCATIONAL INSTITUTIONS

*A web system for teachers to report infrastructure and equipment faults and for administrators to prioritise, resolve and track them.*

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**TECH STACK**

Java 21 · Spring Boot 3.5 · Thymeleaf · Tailwind CSS · MySQL 8 · Flyway · Maven

**DURATION**

3 months

**SUPERVISOR**

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## 1 INTRODUCTION & PROBLEM STATEMENT

### Background

- Schools depend on infrastructure and equipment that must be reported and repaired.
- Manual tracking causes errors and wasted time (Elaskari et al., 2021).

### Problem Statement

- Fault reports are scattered, so status, location and history are unclear.
- Maintenance management falls below best practice (Buys & Nkado, 2006).

## 2 LITERATURE REVIEW

- Asset management is lifecycle-based, not a static list (ISO 55000).
- Barcodes, QR codes and mobile apps automate asset tracking.
- Gap: faults, location and history in one multi-school system.

| Existing work | Strength | Limitation |
|---|---|---|
| Barcode System (Elaskari et al. 2021) | Barcodes cut entry errors | Inventory focus only |
| Equipment Management System using WLAN Technology (Chen et al. 2005) | Centralised across depts | Needs WLAN infrastructure |
| An asset tracking and management system for technical and vocational education (Sulistyo et al. (2022) | QR, history, geolocation | Needs users to scan |
| An inventory management system using Android and QR code technology (Ulum & Sufaidah (2026) | Offline QR inventory | Inventory, not repairs |
| **Our system (Repair Hub)** | Fault reports with status history, for many schools | |

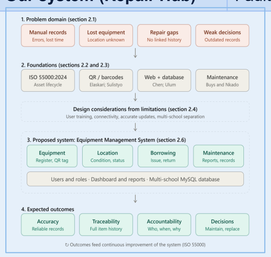



*Figure 1. Conceptual framework*

## 3 SYSTEM DESIGN

- RQ01 Submit reports with priority and a reference number
- RQ02 Attach photos or video as evidence
- RQ03 Record issue location, with GPS on mobile
- RQ04 Teachers track the status of their reports
- RQ05–06 Admins filter reports and update status
- RQ07 Email and in-app notifications
- RQ11 Urgent reports highlighted, unresolved count shown
- Non-functional: HTTPS, 3 s page loads, 50 users

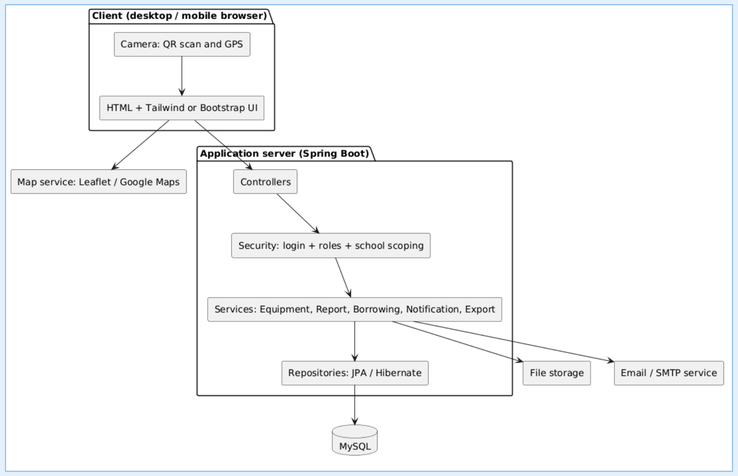



*Figure 2. System architecture*

## 4 RESULTS & DISCUSSION

- Teacher submits a report and receives a reference number.
- Admin dashboard counts unresolved reports and flags priority.
- Status changes are logged with who, when and why.

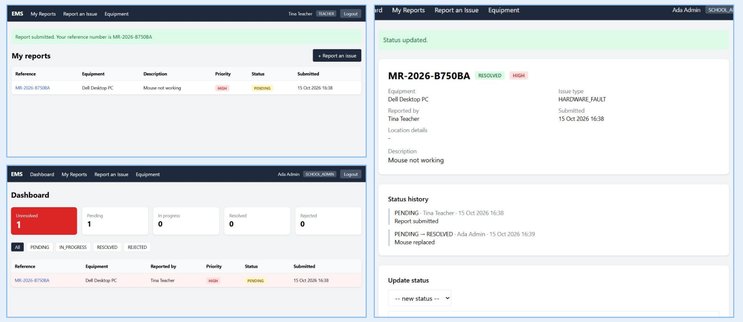



*Figure 3. Working system: teacher report list, admin dashboard, status history*

| Evaluation criterion | Method | Target | Result |
|---|---|---|---|
| Functionality | Test cases per requirement | All pass | [ ] |
| Usability | User testing / SUS survey | [ ] | [ ] |
| Performance | Page load, 50 users | ≤ 3 s | [ ] |

## 5 CONCLUSION

- One central system now records faults, status and history.
- It extends inventory-focused systems with a repair workflow.
- Main requirements: [met / partly met — update].

## 6 RECOMMENDATIONS & FUTURE WORK

- Train teachers and administrators before rollout.
- Limitations: relies on users reporting, and on network access.
- Future: offline mode, RFID tracking and a native mobile app.

## REFERENCES

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